// GLOBEGRID v4 — BUILD MANUAL
// BUILDS ON: v1 + v2 + v3, ALL VERIFIED PRESENT IN THE LIVE REPO

GLOBEGRID v4

POLISH, ACCURACY, AND DEPTH

Fixing what's actually wrong, filling what's actually missing, grounded in the real codebase.

REVISION	4.0
VERIFIED BASE	GlobeGrid repo re-cloned and inspected directly — schema, frontend, and seed data all confirmed against v1-v3 specs
SOURCE OF SCOPE	User's own written reflections on using v3, plus additional items identified during the repo re-read and further ideas
TARGET AGENT	Claude Code / Fable, same CLAUDE.md continuity pattern as v1-v3

Unlike v2 and v3, several items in this manual were verified directly against the live repository rather than taken on description alone — where a bug is described below, the exact code responsible is cited. This matters because it changes some fixes from 'build this' to 'correct this specific, located defect.'

SECTION 01 // VERIFIED STARTING STATE

Scope and verified starting state

The repository was re-cloned fresh and inspected directly (not assumed from prior manuals). Findings:

- `backend/app/db/models.py` confirms every table specified across v1, v2, and v3 exists — including `canonical_entities`, `predictions`, `countries`, `analyst_sessions`, and the full geopolitical entity layer.
- The frontend confirms the v3 components exist as real files: `AnalystPanel.js`, `CountryPanel.js`, `LineageView.js`, `Satellites.js`, `SoundEngine.js`, `TimeScrubber.js`.
- v3 is genuinely complete at the code level. This manual is a refinement and expansion pass, not a rebuild.

1.1 Specific defects confirmed by direct inspection

These are not assumptions — each was located in the actual source before being written up below.

- **Country coverage:** `backend/app/geopolitics/seed_data.py` seeds only ~77 countries, not the ~195 UN members + observers + de facto states the app should represent. Confirms the user's report of Mongolia and Croatia missing — they simply aren't in the file, alongside dozens of others.
- **NATO membership:** the seeded list is `[USA, GBR, FRA, DEU, ITA, CAN, ESP, POL, NLD, TUR, GRC, PRT, NOR, SWE, FIN, CZE, HUN, ROU]` — missing the Baltic states (Lithuania, Latvia, Estonia) and several others (Belgium, Denmark, Iceland, Luxembourg, Slovakia, Slovenia, Croatia, Albania, Montenegro, North Macedonia). Confirms the user's report exactly.
- **ASEAN membership:** the seeded list is `[IDN, THA, VNM, PHL, MYS, SGP, MMR]` — missing Laos, Cambodia, and Brunei. Confirms the user's report exactly.
- **Boundary accuracy:** `frontend/src/data/countryBoundaries.js` is explicitly Natural Earth's 110m (lowest-resolution) dataset — confirms the 'approximates, not accurate lines' complaint at the data-source level, not just a rendering issue.
- **Globe click-through:** `Tier1Globe.js`'s `_hitTest()` rejects far-side points using a loose NDC-depth threshold (`cz/cw > 0.997`) rather than a true camera-facing test — this is a real, locatable cause of the 'orbs clickable from the other side' bug, not a vague rendering quirk.
- **Audio volume:** `SoundEngine.js` sets `this.master.gain.value = 0.06` — roughly 6% of full scale. This is why max system volume is required to hear anything; it's a single-line default, not a deeper audio architecture problem.
- **2D map wraparound:** `Tier2Map.js` contains no wraparound/repeat logic at all — confirms the hard-edge-at-the-Pacific complaint is a genuine gap, not a misconfigured setting.
- **City labeling:** `CITY_LIGHTS` data is used only for the night-side visual glow effect in the globe shader — there is no interactive, zoom-dependent, labeled city layer anywhere in the frontend. The Paradox-style city layer requested doesn't exist yet at all, not just in an unpolished form.
- **Analyst panel context bug:** `state.focusedEntity` in `App.js` is set on every country/location/conflict click but never cleared — not on panel close, not on empty-space click — and the backend's `ask()` gives it unconditional precedence over the actual question text. Full diagnosis in Section 16.

SECTION 02 // GLOBE FIXES

Globe rendering and interaction fixes

2.1 Correct occlusion for clicks (fixes: orbs clickable through the globe)

The vertex shader already computes exactly the right value for this — `vFacing = (mat3(model) * normalize(pos)).z` — but `_hitTest()` and `_hitTestMarked()` don't use it, relying instead on an NDC-depth threshold that doesn't reliably distinguish near-side from far-side points.

- Replace the `cz / cw > 0.997` check in both hit-test functions with the same facing calculation already used for rendering: compute the point's normal in model space, transform by the current model matrix, and reject any candidate whose facing value is negative (facing away from the camera) — not just near the silhouette edge.
- This is a small, precise fix: reuse existing math, don't invent new geometry logic.

2.2 Point clumping and glow obscuring clicks

Two related problems reported together: points bunch up at low zoom, and the bloom/glow effect makes it hard to tell which point you're actually about to click.

- **Zoom-dependent clustering:** at low zoom (globe far away), nearby events collapse into a single cluster marker showing a count — reusing the same clustering concept already specified for the 2D map (v1 Section 5.2) but applied to the 3D globe as well, which doesn't currently have it.
- **Progressive declustering on zoom:** as the camera zooms in, clusters split into their real individual points once they're far enough apart on screen to be individually clickable — not a hard zoom-level cutoff, but a continuous screen-space-distance-based threshold (cluster while screen distance between points is under N pixels, split once it exceeds it).
- **Click-radius independent of glow radius:** decouple the hit-test radius from the visual glow radius — the glow can stay large and pretty, but the actual clickable hit area should be the size of the solid dot underneath it, not the diffuse bloom, so the two don't visually mislead about what's actually clickable.

2.3 Boundary accuracy upgrade

- Replace the 110m Natural Earth boundary set with the same source's higher-resolution 10m dataset (still free, public domain, same provider — no new licensing question) for both the globe and the 2D map (Section 4).
- Given file-size/performance tradeoffs at a glance, use 10m boundaries when zoomed in (accurate coastlines and small-country shapes matter up close) and fall back to a coarser simplification only for the initial full-globe view where sub-kilometer accuracy is invisible anyway — this is the same LOD philosophy as 2.2, applied to boundary geometry instead of points.

SECTION 03 // GEOCODING ACCURACY

Location and geocoding accuracy

The ask is that LLM-driven location pinpointing be as accurate as possible, not an 'uneducated approximation' — this is a real, addressable gap on top of the gazetteer upgrade already shipped in v2.

3.1 Mechanism

- Add an explicit `geocode_confidence` (REAL, 0-1) to `events` — exact gazetteer name match scores highest, fuzzy/alias match scores lower, LLM-inferred-from-context (no explicit place name in the text) scores lowest.
- Low-confidence resolutions are visually distinguished on the map (a dimmer/dashed marker style) rather than presented with the same certainty as a verified exact match — the map should never imply more precision than it actually has.
- For events where the gazetteer can't resolve anything but the text strongly implies a specific place, allow the causal-linker-style LLM call (v1 Section 9's discipline: state uncertainty, don't guess confidently) to propose a candidate location with its own confidence score, gated by the same threshold logic as everything else in this system — never silently upgraded to look verified.

SECTION 04 // 2D MAP OVERHAUL

2D map overhaul

The instruction is direct: the 2D map should be a literal flat version of the globe, not a lesser sibling. Every toggle, every data layer, every accuracy improvement applies to both.

4.1 Feature parity

- Every toggle available on the globe (borders, alliances, non-state actors, border disputes, marked locations/infrastructure, heatmap, ghost trail) gets the identical toggle on the 2D map, reading from the identical backend data — no separate 2D-only data path.
- Country boundaries, alliance tinting, and the 10m boundary upgrade (Section 2.3) apply equally to both renderers.

4.2 Infinite horizontal wraparound

Confirmed gap: `Tier2Map.js` has no wraparound logic today. Top/bottom edges are correct as hard boundaries (there's no wrapping over the poles); only east-west needs to loop.

- Standard technique: render three horizontally-tiled copies of the map (current view, one repeat to the west, one to the east) and pan seamlessly between them, snapping the coordinate space back by 360 degrees of longitude when the pan crosses a tile boundary — the user never perceives a seam.
- All interactive layers (points, borders, marked locations) need to render in all three tiles near the seam, not just the 'primary' one, or markers will appear to vanish right as they'd be most useful (crossing the Pacific).

4.3 Paradox-style city labeling

No such layer exists yet (Section 1.1) — this is a genuine new build, not a polish pass.

- A city dataset with population figures (the same GeoNames source already used for the gazetteer, v2 Section 10, is sufficient — it already has population data per place) drives a population-tiered label system: only the largest cities (e.g. capitals, >5M population) show at low zoom; progressively smaller cities reveal as the user zooms in, exactly the behavior described.
- Label density is screen-space-bounded (don't render more labels than fit without overlapping) rather than a fixed population cutoff per zoom level, so dense regions (e.g. Western Europe) don't overwhelm the view relative to sparse ones.

SECTION 05 // ENTITY DATA COMPLETENESS

Country and entity data completeness

The seed data gap (Section 1.1) isn't a content problem to patch item-by-item — patching in Mongolia, Croatia, and the missing NATO/ASEAN members one at a time just guarantees the next gap surfaces the same way. The fix is a completeness guarantee, not a bigger hand-curated list.

5.1 Mechanism

- Replace the hand-curated `COUNTRIES` list with a full sync against Wikidata's sovereign-state query (already the source used elsewhere in v3, Section 13.2) — this returns the complete UN member/observer list plus commonly-recognized de facto states, not a subset anyone had to remember to include.
- Every alliance's membership list is synced the same way — queried from Wikidata's actual membership data for that organization, not manually transcribed. This is precisely how NATO's Baltic states and ASEAN's Laos/Cambodia/Brunei got dropped in the first place: manual transcription silently omits things.
- Add a startup/CI-style completeness check: compare the seeded country count against a known-correct reference count (e.g. the UN's 193 members + 2 observers) and log a visible warning if the gap is larger than a small tolerance — this turns 'missing countries' from a silent gap into a surfaced, self-reporting problem.

5.2 De facto states and status field

- Add `countries.status`: `un_member | observer_state | de_facto | disputed_territory` — Taiwan, Kosovo, Somaliland, Northern Cyprus, and similar are represented, correctly labeled by status rather than either omitted or presented as uncontroversial.
- Non-state actors (v3 Section 16) already exist as a separate entity type for militant/insurgent groups — de facto *states* stay in the `countries` table with a status flag, since they function as states in practice even without full recognition, which is a different situation from a non-state armed group.

5.3 Border disputes

New table: border_disputes

Field	Type	Description
id	uuid PK	
claimant_a_id	text FK	References countries.id
claimant_b_id	text FK	References countries.id
territory_name	text	
status	text	active | frozen | resolved
boundary_ref	text	Nullable — key into a disputed-boundary geometry set, separate from the main boundary layer
summary	text	

Rendered as its own toggle, off by default alongside standard borders — a disputed line drawn identically to a settled one would misrepresent the actual situation, so disputed boundaries get a visually distinct style (e.g. dashed) whenever the toggle is on.

5.4 Border and non-state-actor visibility toggles

- Standard country borders are visible **by default, always**, with a single settings toggle to turn them off entirely — this reverses the current implicit behavior into an explicit, guaranteed-on default per the user's instruction.
- Non-state actors (v3 Section 16) get their own visibility toggle, independent of the border toggle and independent of the border-disputes toggle (Section 5.3) — three separate, orthogonal switches, not one overloaded 'extra layers' toggle.

5.5 Universal profile coverage

Every country, every de facto state, and every registered non-state actor must have a navigable profile page — this is a completeness requirement on the profile system (v3 Sections 13 and 16), not a new mechanism. The startup completeness check (5.1) should also verify that every seeded entity actually has a resolvable profile record, not just a map boundary.

SECTION 06 // PROFILE UI & WIKI

Country profile UI redesign and internal wiki system

6.1 Country profile panel redesign

- Move from whatever ad hoc placement exists today to a fixed **left-side docked panel** — formal, structured, consistent chrome (matches the analyst panel's docked-right convention from v3 Section 24, giving the app a coherent 'two docked panels' visual language rather than floating windows).
- Header block: country flag, a photo of the current head of state/government (sourced alongside the existing Wikidata leadership sync, v3 Section 13.2 — Wikidata entries for heads of state typically include a linked image), and the ruling party/coalition name, all visible without scrolling.
- Below the header: the existing profile sections (legislature, agendas, market/trade, stances) in a cleaner, more formal layout than the current implementation — this is a visual/CSS pass on existing data, not new backend work.

6.2 Internal wiki system for parties, leaders, and other entities

Extends the profile concept to notable persons (v3 Section 22), political parties (new), and other entity types, presented consistently as wiki-style pages rather than one-off layouts per entity type.

- New table: `political_parties` (id, name, country_id FK, ideology_tags, founded_date, canonical_entity_id FK) — parties become first-class linkable entities, the same pattern used for non-state actors and international organizations.
- Every wiki-style page (country, party, leader, non-state actor, international organization) supports **previous/next navigation arrows** — browsing sequentially through, e.g., every notable person, without returning to a directory each time.
- A **fullscreen toggle** expands the currently-open page to take up the majority of the map/globe viewport; **minimized** (docked to the left pane per 6.1) is the default state.

SECTION 07 // EXTERNAL CONTENT SOURCING

Generating profile content from external sources

Country/party/leader summaries and national-agenda narratives (e.g. China's Belt and Road / New Silk Road) should draw on established web sources, not only GlobeGrid's own ingested story stream — background knowledge and current events are different kinds of input and both matter here.

7.1 Sources

- **Wikipedia:** stable, well-structured, free, no API key required (MediaWiki API) — the primary background-knowledge source for a profile's static/historical content.
- **Grokipedia:** verified directly before writing this — Grokipedia (xAI's AI-generated encyclopedia, live at grokipedia.com) currently operates as a **read-only web platform with no official public API**. An unofficial third-party Python client exists (`grokipedia-api` on PyPI) that wraps the read-only interface, but it is community-maintained, not xAI-sanctioned, and could break without notice. Flagged **OPEN**: integrate it as a best-effort secondary source behind its own feature flag, never a required dependency, and re-check its status before relying on it further — this is a case where the honest answer is 'not yet stable enough to be load-bearing,' not a reason to skip it entirely.
- **Reliable news coverage**: reuses the existing ingestion pipeline (v1 Section 4, v2 Section 1) rather than a new mechanism — a country's agenda synthesis (v3 Section 13.2) already pulls from tagged stories; this just means ensuring that pipeline's source breadth (Section 13 below) is wide enough to catch agenda-relevant coverage specifically.

7.2 Mechanism

- Background synthesis (Wikipedia + Grokipedia-if-available) runs on the same slow refresh cadence as other reference data (v3 Section 13.2's weekly Wikidata sync) — this is scene-setting content, not a live feed.
- Current-events synthesis continues to come from GlobeGrid's own tagged story stream, exactly as v3 already specifies — the two source types are blended in the profile's presentation but kept clearly attributed by origin (background source vs. tracked coverage), never merged into a single unsourced paragraph.

SECTION 08 // STORY TAXONOMY & BROWSING

Broader story types and story browsing

The current notion of a 'story' (a correlated cluster of events, v1 Section 5.4) is real but narrow. A war, a developing alliance, a repeating historical pattern of conflict, and a country's diplomatic or economic push are all 'stories' in the sense the user means, but only some of them look like a tight cluster of recent events.

8.1 Story type taxonomy

New column: stories.story_type

Value	What it's backed by
`acute_event`	The existing default — a tight cluster of recent correlated events (unchanged behavior)
`ongoing_conflict`	Backed by the conflicts entity (v3 Section 15) rather than a single cluster — long-running, continuously updated
`alliance_development`	A bloc's membership, posture, or agreements changing over time — backed by alliances (v3 Section 14) and bilateral_relations (v3 Section 19)
`recurring_pattern`	A historical pattern surfaced by the lineage view (v3 Section 8) or second-order causal reasoning (v3 Section 3.7) — repeated conflict cycles, recurring diplomatic breakdowns
`diplomatic_push`	A country's sustained agenda toward a goal — backed by country_agenda_synthesis (v3 Section 13.2)
`economic_push`	Same as above, backed by the economic_agenda field specifically, plus sanctions/treaties (v3 Sections 20-21) where relevant

8.2 Story browsing UI

- A dedicated 'Stories' directory tab, filterable by `story_type`, separate from the live feed — the live feed stays chronological/recency-driven (v1 Section 5.1), while this directory is the browsing surface for the slower-moving story types that don't naturally rise to the top of a recency-sorted feed.
- Each story type gets presentation appropriate to its shape — an acute_event still looks like today's story card, but an alliance_development or diplomatic_push renders more like a timeline/narrative arc than a single event cluster.

SECTION 09 // FILTERING & RELEVANCE

Filtering and relevance

9.1 Local-news relevance filter

Problem: routine local coverage (a local police blotter item, a minor municipal dispute) is noise for a global-geopolitics tool, but it isn't always obviously distinguishable by source alone — some outlets mix wire-level and hyper-local coverage in the same feed.

- Add a `events.global_relevance_score` (REAL, 0-1), computed at extraction time from a small set of signals: does the event have any canonical-entity link to a country/organization/conflict (v2 Section 3.1, v3 Sections 13-23) beyond just a place name; does its severity (v1 Section 6.3) clear a minimum floor; is the originating source itself typically global vs. hyper-local in scope.
- A single relevance-floor toggle in settings ('Hide low-relevance local coverage') filters the live feed and map by this score — off by default is debatable, but given the tool's explicit purpose, defaulting **on** (filtering out low-relevance noise) fits the product better than showing everything and asking the user to opt into a cleaner view.

9.2 Continent and conflict filters

- Continent filter: a straightforward filter on `countries.region` (already present, v3 Section 13.2's schema) applied across feed, map, and stories directory (Section 8.2) simultaneously — one filter state shared across all three views, not three separate settings that can drift out of sync.
- Conflict filter: reuses the existing per-conflict tab mechanism (v3 Section 15.1) directly — 'windows for specific conflicts' is what that section already specifies; this item confirms it should be reachable as a filter from the main feed/map, not only via the conflicts directory.

SECTION 10 // EXPERT PROGNOSIS

Expert prognosis on conflicts and diplomatic processes

This extends the existing forecasting feature (v3 Section 7) — which was deliberately scoped to short-horizon risk statements — to also cover the slower, more analytical question of where an ongoing conflict or diplomatic process seems to be heading. Same discipline applies: routed through the tracked, graded prediction pathway, never a free-floating opinion.

10.1 Mechanism

- A conflict or diplomatic-push story (Section 8.1's `ongoing_conflict` / `diplomatic_push` types) can carry a periodically-refreshed 'prognosis' — a longer-horizon, more discursive companion to the short-horizon forecast already specified in v3 Section 7, using the same `predictions` table with `kind = forward_forecast` and a longer `horizon_hours` value rather than a separate mechanism.
- Same confidence-capping and track-record-display requirements from v3 Section 7.2 apply without exception — a longer time horizon does not earn a pass on the same discipline that governs the 72-hour forecasts.

SECTION 11 // TEXT QUALITY

Text normalization and deeper summaries

11.1 Title/summary normalization pass

Ingested headlines vary wildly in capitalization convention, punctuation, and occasional grammar/spelling issues (especially from non-native-English or wire-service sources). A normalization pass makes the feed read consistently without altering meaning.

- Applied at display-generation time (when a story's headline/summary is synthesized, v1 Section 5.5), not to the raw stored source text — the original always remains unmodified and available for exact citation; only the system's own generated headline/summary text gets normalized.
- A lightweight, deterministic pass (standard title-case rules, basic punctuation cleanup) handles the bulk of cases; only genuinely malformed input needs a model call, keeping this cheap.

11.2 Deeper summaries

- Increase the target length/detail of the AI-generated per-story summary (v1 Section 5.3) — the current one-paragraph summary is a floor, not a ceiling; a 'read more' expansion revealing a fuller synthesis (still source-linked, per the attribution requirement running through this entire project) gives depth on demand without cluttering the default card view.

SECTION 12 // AUDIO SYSTEM

Audio fixes and generative music expansion

12.1 Volume fix

Confirmed root cause (Section 1.1): `this.master.gain.value = 0.06` in `SoundEngine.js`. Raise the default to a genuinely audible level (a starting point around 0.35-0.45 is reasonable for an ambient layer that shouldn't dominate, versus the current ~6% of full scale) and expose a proper in-app volume slider so the user sets it once rather than fighting system volume.

12.2 Additional generative music presets

Extends the existing generative music system (v3 Section 9.2 — instability-driven key/tempo, regional instrument voices) with selectable genre presets, each swapping the oscillator/scale/rhythm character while keeping the same instability-reactive driving logic underneath.

New presets

Preset	Character
Technofuturistic / vaporwave	Slower sidechain-style pulse, chorused pads, pitched-down sub bass, a nod to the aesthetic the whole app already leans into
Dune-inspired	Sustained analog-style synth drones with high, strong string-like lines layered in during high-instability periods — movement, conflict, and hope in the same texture, as described
Ambient metal / hard rock	Detuned power-chord-style low drones with occasional distorted transient hits timed to burst events (v2 Section 5.1's story-formation burst)
Technohouse	Four-on-the-floor pulse tempo-locked to the instability score, arpeggiated sequences over the existing regional-voice panning

- Selectable in settings alongside the existing mute/volume controls; the underlying instability-reactive mapping (v3 Section 9.2) is preset-agnostic, so adding more presets later is a matter of defining a new scale/timbre profile, not new architecture.

12.3 Further experimental presets

Going further, on-theme with the intelligence-briefing aesthetic this whole project already leans into (the build manuals themselves use a 'classified internal spec' framing) — a few genuinely different textures, not just variations on the same electronic palette as 12.2.

More presets

Preset	Character
Numbers station	Shortwave static bed, sparse detuned piano notes, occasional morse-code-like blip sequences keyed to new story arrivals — leans directly into the classified-intelligence-broadcast aesthetic already used in this project's own documentation
Berlin industrial	Harder and colder than the existing technohouse preset — mechanical, slightly distorted percussion, minimal melodic content, four-on-the-floor but stripped down; the most literally 'techno' of any preset in either theme
War-room orchestral	Low brass swells and timpani hits timed to severity spikes, sparse string underlay — a news-documentary underscore rather than an electronic track; the only fully acoustic-modeled preset, a real tonal departure from everything else
Data sonification / glitch	Doesn't just react to the instability score — sonifies the live event stream itself, generating a distinct granular/glitch texture per incoming event_created message (Section 2 of this manual), so the sound is a direct audible representation of ingestion happening, not just a mood layer
Modal drift	Shifts toward different modal/microtonal drone scales based on which region currently has the most active events — implemented as abstract scale/interval choices rather than literal regional instrument samples, deliberately avoiding caricature while still giving the sound a sense of place
Arctic / calm state	Very slow, glacial, minimal — the deliberate low-energy counterpart to every other preset; quiet periods should sound genuinely calm, not just a quieter version of the tension music

The data-sonification preset needs slightly different plumbing than the others — it subscribes to the WebSocket event stream directly (v1 Section 8.2 / Section 2 of this manual) rather than only reading the periodically-recomputed instability score, so it belongs in its own small implementation pass rather than being dropped in alongside the others as a pure preset swap.

SECTION 13 // INGESTION BREADTH & SPEED

Ingestion breadth, speed, and market coverage

The ask is for GlobeGrid to have 'sufficient context on what is going on in all the conflicts' — this is explicitly about backpropagation depth (how far back the system's context reaches) and breadth (how many countries/currents are actually covered), not just faster polling.

13.1 Mechanism

- Widen source coverage per-country: ensure every seeded country (Section 5.1, now the full ~195) has at least one relevant tracked source or is covered by a broad-scope source (GDELT, v1 Section 4) — the current source list was built around a handful of major outlets, which structurally under-covers most of the world.
- Backpropagation: on first enabling a country/conflict profile, run a one-time historical backfill query against GDELT's rolling window (v1 Section 4.2) and, where available, Wikipedia's own article history for that topic, rather than only starting to accumulate context from the moment the entity was added — this directly serves the 'sufficient context on what's going on' requirement, since a freshly-added conflict shouldn't start with zero history.
- Increase ingestion frequency for high-activity conflicts specifically (dynamic polling interval scaled by recent event volume for that conflict_id, v3 Section 15) rather than a uniform interval for everything — an actively escalating conflict deserves tighter polling than a frozen one.

13.2 Economic and market story gathering

- Widen the market/economic source set beyond the existing Alpha Vantage integration (v1 Section 4.4, v2 Section 1.4) — add coverage of central-bank statements and major economic-data-release calendars as their own tracked sources, since these are structurally different from both news articles and raw price ticks and currently fall into neither bucket well.
- Tag economic events more specifically at extraction time (v1 Section 5.1) so they route cleanly into a country's economic_agenda synthesis (v3 Section 13.2) rather than being caught only incidentally by general news coverage.

SECTION 14 // API KEY ONBOARDING

In-app API key setup guide

A real onboarding page, not just documentation — the app should be usable by someone who has never touched a `.env` file.

14.1 Mechanism

- A first-run setup screen (or a persistent Settings > API Keys page) walks through obtaining and entering the Claude API key specifically — where to sign up, where the key appears on Anthropic's console, and a paste field that writes it to the local `.env` file directly rather than requiring manual file editing.
- The same page covers the other optional keys already in the stack (Alpha Vantage, Reddit) with the same guided pattern, and clearly marks which are required (Claude) versus optional (everything else degrades gracefully without them, consistent with the zero-install philosophy already established).
- A visible status indicator confirms each key is actually working (a lightweight test call) rather than just accepted and hoped-for.

SECTION 15 // ADDITIONAL IDEAS

Further ideas beyond the reflections list

A few things worth adding that weren't explicitly requested but follow naturally from everything above.

15.1 Coverage/confidence transparency

- A visible 'thin coverage' indicator on any country/conflict profile whose backing story count falls below a threshold — the honest alternative to presenting sparse data with the same confidence as well-covered topics. Directly extends the geocode-confidence idea (Section 3.1) to the entity-profile level.

15.2 Accessibility

- A colorblind-safe palette toggle — the system leans heavily on color for severity/instability/alliance-tinting, which is a real accessibility gap worth closing given how central color coding is to the whole UI.

15.3 Visible freshness

- Every profile (country, conflict, alliance, non-state actor) displays its own 'last updated' timestamp plainly — ties directly to the 'updated in real time' requirement from the original ask: the honest version of real-time is showing exactly how fresh the data actually is, not just asserting freshness.

15.4 First-launch performance calibration

- Since Tier detection (v1 Section 11) is capability-based but the new zoom-dependent LOD systems (Sections 2.2, 4.3) are genuinely new rendering load, a brief one-time calibration pass on first launch (render a representative frame, measure actual frame time) tunes default cluster/label density thresholds to the specific machine rather than a single hardcoded default for everyone.

SECTION 16 // ANALYST PANEL FIX

Diagnosing and fixing the analyst panel

Corrected diagnosis: the symptom isn't content defaulting to a specific answer — it's that the panel keeps answering in the context of whatever page was last opened, regardless of the new question. Confirmed by tracing the actual click-to-focus wiring end to end.

16.1 Root cause, confirmed

`App.js` sets `state.focusedEntity` every time a country, location, or conflict is clicked (lines setting it on country click, location click, and conflict selection) — but **nothing ever clears it back to null**. `CountryPanel.close()` only hides the panel's DOM; it never touches `state.focusedEntity`. Worse, the conflict-selection line reads `state.focusedEntity = active ? {...} : state.focusedEntity` — when nothing is actively selected, it explicitly *keeps the previous stale value* rather than clearing it.

On the backend, `ask()` checks `focused_entity` first and unconditionally — if it resolves to anything, that becomes the answer's context, and the actual question text is never even checked against the entity layer. Combine the two: click any country once, close the panel, and every subsequent question for the rest of the session answers in that country's context — exactly the reported behavior.

16.2 Fix

- **Frontend:** clear `state.focusedEntity = null` explicitly whenever a panel is closed (add this to `CountryPanel.close()` and the equivalent for story/wiki pages), when the user clicks empty space on the globe/map, and change the conflict-selection line to set `null` when nothing is active instead of preserving the old value.
- **Backend:** stop giving `focused_entity` unconditional precedence. Always attempt `_entity_match(question)` against the question text itself first; only fall back to the focused-entity context when the question's own text resolves to nothing. A stale or even a genuinely-current focus should never override what the user is actually asking.
- **Defense in depth:** also timestamp `focusedEntity` when it's set, and ignore it entirely if it's older than a short window (e.g. 2-3 minutes) — so even if a future code path forgets to clear it explicitly, staleness self-corrects.

Separately worth knowing, unrelated to this bug: whether `CLAUDE_API_KEY` is set still determines whether answers get full AI synthesis or the shorter retrieval-only summary (`backend/app/config.py`) — worth checking as a separate, independent thing, but it is not the cause of this specific bug.

16.3 De facto states, still worth fixing

While tracing this, also confirmed Palestine is absent from the `countries` seed data entirely, and that the conflict-name fuzzy matcher requires 2 of 3 tokens for multi-word conflict names (so 'Palestine' alone wouldn't match `Israel–Palestine Conflict` even with the focus bug fixed). Both remain real, worth fixing alongside the above — the first is already covered by Section 5.2; the second needs its own small fix: strip generic trailing words (`war, conflict, crisis, dispute`) before applying the token threshold, and register each side's name as an independent alias.

SECTION 17 // EVENT CARD PANE

Event cards in the sliding pane

Currently, clicking a country orb opens the country profile in the left-docked panel (v3/v4 Section 6.1), but clicking an event orb opens its story page some other way — an inconsistency worth closing.

17.1 Mechanism

- Story pages (v1 Section 5.3) render in the identical left-docked sliding panel used for country/wiki pages (Section 6.1-6.2), not a separate window or modal — one consistent panel component serves country profiles, wiki pages, and event/story cards alike, distinguished only by content template.
- The panel's navigation history (previous/next, Section 6.2) applies here too: opening an event from the feed, then opening a linked country from within that event's story, then hitting 'back' returns to the event — a real navigation stack, not independent one-off panels.

SECTION 18 // PANE ANIMATION SYSTEM

Fluid animation across all panes

A single shared animation system, not per-component ad hoc transitions — this is what makes 'all panes feel cool and techno' actually consistent rather than each panel having its own slightly-different motion feel.

18.1 Mechanism

- One shared CSS/JS transition module used by every docked panel (country/wiki panel, analyst panel, event cards, settings) — slide-in from the dock edge with an eased cubic-bezier curve (fast start, gentle settle), consistent duration (~280-320ms) across all of them.
- Content *within* a panel cross-fades on navigation (previous/next, Section 6.2) rather than hard-cutting — the panel shell stays in place and slides, the content inside fades/slides a shorter distance, so switching pages feels like turning a page, not reloading a window.
- Fullscreen toggle (Section 6.2) animates the expansion/contraction itself (panel width and content reflow tween together) rather than snapping between two fixed states.
- Respect `prefers-reduced-motion` — disable/shorten these transitions for anyone with that OS-level setting, small but easy to get right while building this.

SECTION 19 // PERSONAL ANNOTATIONS

Personal annotations layer

Not explicitly requested, but a natural fit: this is explicitly a single-user, personal analytical tool, and the wiki-style browsing system (Section 6.2) plus the AI's own causal reasoning (v1 Section 9) invites the user's own layer of analysis alongside it — a place to record your own read on a story, separate from and visually distinct from the AI's.

New table: annotations

Field	Type	Description
id	uuid PK	
target_type	text	story | country | conflict | non_state_actor | alliance
target_id	text	Matches the id of whatever entity type this annotates
note_text	text	
created_at	timestamptz	
updated_at	timestamptz	

- Rendered as a distinctly-styled panel section (different from AI-generated content, never blended with it) on any profile or story page — your own notes are clearly yours, not mistaken for tracked/sourced system output.

SECTION 20 // REASONING TRACE

Correlation reasoning trace view

Given how much weight this whole project puts on the correlation engine being the centerpiece, a 'show your work' view is a natural next step: an expandable trace on any story showing exactly why the correlation engine grouped these particular facts together — which threshold fired, what the actual similarity score was, whether it was a same-window or historical-chain link (v1 Section 5.4).

- Surfaces data that's already computed and stored (similarity scores, `linked_via` on story_members, v1 Section 6.6) rather than requiring new computation — this is a transparency/presentation feature, not a new analytical capability.
- For debate-enabled stories (v3 Section 2), the trace can also show the three-persona disagreement score alongside the correlation trace — one place to see everything the system 'thought' about a story, not scattered across separate views.

SECTION 21 // BOOKMARKS

Bookmarks and reading list

The wiki-style browsing system (Section 6.2) with prev/next navigation implies exactly the kind of reference-browsing behavior where 'save this for later' is a natural need — distinct from watchlists (v2 Section 6.2), which are about ongoing monitoring rather than personal curation of things already read.

New table: bookmarks

Field	Type	Description
id	uuid PK	
target_type	text	story | country | conflict | non_state_actor | alliance | notable_person
target_id	text	
bookmarked_at	timestamptz	

- A single bookmark icon on every profile/wiki/story page (consistent placement, same component everywhere per Section 18's shared-system philosophy) and a dedicated 'Bookmarks' directory to revisit them.

SECTION 22 // SOURCES & CREDITS

Sources and credits page

Attribution to individual stories has been a hard requirement since the very first version of this project (v1 Section 6.8). A single consolidated credits page is the natural extension of that same ethos to the data providers themselves — GDELT, Wikidata, GeoNames (with its required CC BY 4.0 credit, v2 Section 10.3), USGS, NASA FIRMS, Alpha Vantage, Reddit, Wikipedia, and any others — especially relevant for a project headed toward free public release.

- A static, always-accessible page (not generated per-session) listing every registered source (v1 Section 6.1) grouped by category, with attribution text/links pulled from a small per-source-type metadata field rather than hardcoded prose, so adding a new source type doesn't require a manual credits-page edit.

SECTION 23 // SNAPSHOT EXPORT

Shareable snapshot export

v3's timelapse export (Section 10.1) is a full video; this is the lighter, faster sibling — a single beautiful still image of the current globe view plus a short text overlay of top stories, sized for sharing. Worth having given the eventual TalkDiplomacy integration and the Instagram-adjacent framing raised earlier in this project's planning.

- Client-side canvas capture of the current Tier 1 render at a fixed export resolution (e.g. 1080x1080 for square social formats), composited with a text overlay (top 2-3 story headlines, instability score, timestamp) and the attribution footer required by Section 22.
- One click, no server round-trip — this should be as close to instant as the timelapse export (v3 Section 10.1) deliberately isn't.

SECTION 24 // COMPARE VIEW

Side-by-side compare view

A natural extension once bilateral relations (v3 Section 19), country profiles (v3 Section 13), and the wiki-style panel (Section 6) all exist: let the user pick any two countries, conflicts, or alliances and see them rendered side by side in split panels, rather than only being able to view one entity at a time.

- Reuses the exact same profile/wiki content templates from Section 6 in a two-column layout — no new content generation, purely a presentation mode.
- For two countries specifically, the relevant bilateral_relations row (v3 Section 19) surfaces prominently between the two columns, since that's usually the actual reason someone wants to compare them.

SECTION 25 // IN-APP ALERTS

In-app breaking story alerts

Distinct from the 'get it off the screen' category explicitly excluded from v3 (OS tray notifications, smart-home lighting) — this stays entirely on-screen, a brief animated toast within the app itself when a high-severity story forms while the user's attention is elsewhere in the UI (e.g. deep in a country profile, Section 6).

- Triggered by the same `story_created` WebSocket event already broadcast today (v1 Section 8.2), filtered to a severity floor (config-driven) so routine stories don't generate constant interruptions.
- Uses the shared animation system (Section 18) for its own slide-in/fade — consistent motion language, not a one-off toast implementation.
- Clicking the alert opens that story directly in the sliding pane (Section 17) — the alert is a shortcut into the existing system, not a separate notification UI with its own content rendering.

SECTION 26 // CONFIG ADDITIONS

New config.yaml values

Appended to the config file as it stands after Sections 2-25 above — nothing prior is removed.

```yaml
globe:
  hit_test_use_facing_occlusion: true   # replaces the old NDC-depth threshold
  cluster_screen_distance_px: 40
  boundary_resolution: "10m"            # was 110m
  boundary_resolution_far_zoom: "50m"   # coarser fallback for full-globe view

geocoding:
  min_confidence_for_solid_marker: 0.6

map2d:
  wraparound_enabled: true
  city_label_min_population_by_zoom:
    far: 5000000
    mid: 500000
    near: 50000

entity_completeness:
  reference_country_count: 195
  completeness_warning_tolerance: 5

relevance:
  global_relevance_default_filter: true
  global_relevance_floor: 0.3

audio:
  master_gain_default: 0.4              # was 0.06
  preset: "ambient_default"             # ambient_default | vaporwave | dune | metal | technohouse
                                         # | numbers_station | berlin_industrial | war_room_orchestral
                                         # | data_sonification | modal_drift | arctic_calm

sourcing:
  historical_backfill_on_entity_add: true
  dynamic_polling_by_conflict_activity: true

external_content:
  wikipedia_enabled: true
  grokipedia_enabled: false             # OPEN -- unofficial client only, off by default until re-verified

onboarding:
  require_claude_key_before_first_run: true

analyst_panel_fixes:
  clear_focus_on_panel_close: true
  focus_staleness_timeout_seconds: 180
  question_text_takes_precedence_over_focus: true
  conflict_token_strip_generic_words: true
  conflict_alias_individual_parties: true

panes:
  transition_duration_ms: 300
```

```
    respect_prefers_reduced_motion: true

  annotations:
    enabled: true

  alerts:
    in_app_breaking_alert_severity_floor: 4
```

SECTION 27 // BUILD SEQUENCE

Build sequence for Claude Code

Everything above is in scope. Sequencing groups the confirmed, precisely-located bug fixes first (cheapest, highest-confidence wins), then the data-completeness fix that several other items depend on, then the larger UI/content builds.

v4 phases

Field	Type	Description
Phase BB	Confirmed bug fixes	Sections 2.1 and 12.1 -- occlusion hit-test fix and audio gain fix; both are precisely located, one-file changes, do these first
Phase CC	Entity data completeness	Section 5 -- the Wikidata-driven country/alliance sync and completeness check; several later phases (profile coverage, filters, story types) assume this is done
Phase DD	Globe/map LOD and boundary upgrade	Sections 2.2, 2.3, 4 -- clustering, boundary resolution, 2D map parity and wraparound, city labeling
Phase EE	Geocoding confidence	Section 3 -- depends on Phase CC's entity layer being complete enough to be a meaningful comparison baseline
Phase FF	Profile UI redesign + internal wiki	Section 6 -- depends on Phase CC; political_parties is a new table introduced here
Phase GG	External content sourcing	Section 7 -- depends on Phase FF existing to display the synthesized content into
Phase HH	Story taxonomy + browsing directory	Section 8 -- depends on Phase CC (alliances/conflicts/agendas all need to exist to back the new story types)
Phase II	Filtering and relevance	Section 9 -- depends on Phase CC (continent filter) and existing conflict tabs (v3 Section 15)
Phase JJ	Expert prognosis	Section 10 -- extends v3's forecasting; same sequencing caution as v3 Section 7 applies (needs scorecard history)
Phase KK	Text quality pass	Section 11 -- independent, can run anytime
Phase LL	Audio presets	Section 12.2 -- independent, cheap, can run anytime after Phase BB's gain fix
Phase MM	Ingestion breadth and market coverage	Section 13 -- independent, ongoing work rather than a one-time build
Phase NN	API key onboarding page	Section 14 -- independent, worth doing early since it affects first-run experience
Phase OO	Additional ideas	Section 15 -- small, independent items; fold into whichever phase touches the same UI surface
Phase PP	Analyst panel fixes	Section 16 -- do first among this batch: clear stale focusedEntity on panel close/empty click, and stop giving it unconditional precedence over the question text; both are precisely located
Phase QQ	Event card pane + universal animation	Sections 17-18 -- build together, the animation system should exist before wiring the event-card pane into it
Phase RR	Personal annotations + bookmarks	Sections 19, 21 -- independent, small, share a similar schema shape
Phase SS	Reasoning trace view	Section 20 -- depends on Phase QQ's pane system to render into

Field	Type	Description
Phase TT	Sources/credits page + snapshot export	Sections 22-23 -- independent, both relatively small and self-contained
Phase UU	Compare view + in-app alerts	Sections 24-25 -- depend on Phase QQ's pane and animation system

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// END OF MANUAL -- GLOBEGRID v4 BUILD MANUAL -- REV 4.0